

12/4/20

Measure of Central Tendency

I Mean

Q1 Define the meaning of central Tendency.
 Ans Central tendency is the single value which is used to represent the entire set of data. The most commonly average are mean, median and mode.

Q2 Define Arithmetic Mean.

Ans It is the most simplest and commonly used measure of central Tendency. It is commonly denoted by, $\bar{x}$, there are three ways of calculating mean.

- * Direct method
- * Short cut method
- * Step Deviation method

Ex-8-Sol

x	f	fx	$d = x - A$	fd	$d' = \frac{d}{c}$	fd'
2	5	10	-3	-15	-3	-15
3	8	24	-2	-16	-2	-16
4	9	36	-1	-9	-1	-9
$\Sigma f = A$	8	40	0	0	0	0
6	16	96	1	16	1	16
7	10	70	2	20	2	20
8	17	136	3	51	3	51
9	12	108	4	48	4	48
$\Sigma f = 80$		$\Sigma fx = 480$	$\Sigma fd = 80$		$\Sigma fd' = 80$	

Direct Method:

$$(\bar{x}) = \frac{\Sigma fx}{\Sigma f} = \frac{480}{80} = 6$$

Short cut Method:

$$(\bar{x}) = A + \frac{\Sigma fd}{\Sigma f} = 5 + \frac{80}{80} = 6$$

Step-Deviation Method:

$$(\bar{x}) = A + \frac{\Sigma fd'}{\Sigma f} \times c = 5 + \frac{80}{80} \times 1 = 6$$

Ex-12-Sol

						$C=10$	
x	f	m	fm	$d=m-A$	fd	$d'=\frac{d}{C}$	fd'
0-10	10	5	50	-40	-400	-4	-40
10-20	8	15	120	-30	-240	-3	-24
20-30	12	25	300	-20	-240	-2	-24
30-40	16	35	560	-10	-160	-1	-16
40-50	20	$A=45$	900	0	0	0	0
50-60	17	55	935	10	170	1	17
60-70	14	65	910	20	280	2	28
70-80	8	75	600	30	240	3	24
$\Sigma f = 100$		$\Sigma fm = 4100$	$\Sigma fd = -400$		$\Sigma fd' = -40$		

Q-21-22

x	f	fx
20	1	20
35	2	70
40	3	120
x	4	4x
50	6	300
75	4	300
$\Sigma f = 20$		$\Sigma fx = 820 + 4x$

$$(\bar{x}) = \frac{\Sigma fx}{\Sigma f} = \frac{820 + 4x}{20} = 50$$

$$1000 = 820 + 4x$$

$$4x = 180$$

$$x = 45$$

Ex-22-51

x	f	fx
5	5	25
15	7	105
25	x	25x
35	18	630
45	8	360
55	3	165
$\Sigma f = 48 + x$		$\Sigma fx = 1150 + 25x$

$$(\bar{x}) = \frac{\Sigma fx}{\Sigma f}$$

$$29 = \frac{1150 + 25x}{48 + x}$$

$$1102 + 29x = 1150 + 25x$$

$$4x = 48$$

$$x = 12$$

Revision Exercise - Missing value

22-50

x	f	fx
20	8	160
30	12	360
40	20	800
x	10	10x
60	6	360
70	4	280
$\Sigma f = 60$		$\Sigma fx = 1960 + 10x$

$$(\bar{x}) = \frac{\Sigma fx}{\Sigma f}$$

$$41 = \frac{1960 + 10x}{60}$$

$$2460 = 1960 + 10x$$

$$500 = 10x$$

$$x = 50$$

23-50

x	m	f	fm
10-20	15	17	255
20-30	25	18	450
30-40	35	10	350
40-50	45	x	45x
50-60	55	2	110
$\Sigma f = 47 + x$		$\Sigma fm = 1165 + 45x$	

$$\bar{x} = \frac{\sum fm}{\sum f}$$

$$76 = \frac{1165 + 45x}{47 + x}$$

$$1112 + 25x = 1165 + 45x$$

$$57 = 19x$$

$$x = 3$$

x	m	f	fm
0-10	5	12	60
10-20	15	2	15x
20-30	25	17	425
30-40	35	7	35y
40-50	45	17	765
50-60	55	6	330

$$\sum f = 61 + x + y = 100 \quad \sum fm = 1930 + 15x + 35y$$

$$\therefore x + y = 39$$

$$65 = \frac{\sum fm}{\sum f}$$

$$78 = \frac{1930 + 15x + 35y}{100}$$

$$7800 = 1930 + 15x + 35y$$

$$5870 = 15x + 35y$$

$$194 = 3x + 7y$$

Test Yourself 8.6

1-solu $\bar{x}_{1,2} = 72$
 $\bar{x}_1 = 76$
 $\bar{x}_2 = 66$

$$\bar{x}_{1,2} = \frac{N_1 \bar{x}_1 + N_2 \bar{x}_2}{N_1 + N_2}$$

$$72 = \frac{N_1(76) + N_2(66)}{N_1 + N_2}$$

$$72N_1 + 72N_2 = 76N_1 + 66N_2$$

$$6N_2 = 4N_1$$

$$3N_2 = 2N_1$$

$$3N_2 = N_1$$

$$2$$

$$\frac{3(100-x)}{2} = x \quad (N_1 = x; N_2 = 100-x)$$

$$2$$

$$\frac{300 - 3x}{2} = x$$

$$2$$

$$300 = 2x + 3x$$

$$x = 60$$

$$100 - x = 100 - 60$$

$$= 40$$

$$\text{Boys} = 60\%$$

$$\text{Girls} = 40\%$$

2-200 $\bar{x}_1 = 162 \text{ cm}$

$N_{1,2} = 200$

$N_1 = 125$

$\bar{x}_1 = 165 \text{ cm}$

$$\therefore 162 = \frac{125(165) + (200-125)(\bar{x}_2)}{125 + (200-125)}$$

$$162 = \frac{125(165) + 75\bar{x}_2}{125 + 75}$$

$$162 \times 200 = 125(165) + 75\bar{x}_2$$

$$32400 = 20625 + 75\bar{x}_2$$

$$75\bar{x}_2 = 11775$$

$$\bar{x}_2 = \frac{11775}{75} = 157$$

3-300 $N_{1,2} = 40$

$(\bar{x}_1) = 1,00,000$

x	f	fx
80,000	5	4,00,000
90,000	8	7,20,000
1,00,000	6	6,00,000
1,10,000	7	7,70,000
1,20,000	4	4,80,000
$\Sigma f = 20$		$\Sigma fx = 19,70,000$

$$(\bar{x}) = \frac{19,70,000}{20} = 99,000$$

$$\therefore 1,00,000 = \frac{30(99,000) + 10(\bar{x}_2)}{40}$$

$$40,00,000 = 29,70,000 + 10\bar{x}_2$$

$$10\bar{x}_2 = 10,30,000$$

$$\bar{x}_2 = 1,03,000$$

Test Yourself 8.7

1-Solo Wrong $(\bar{x}) = 20$

$N = 20$

Incorrect value = 22

Correct value = 22

$$\therefore \text{Correct } (\bar{x}) = \frac{20 - 22 + 32}{20} = \frac{30 + 10}{20} = \frac{40}{20} = 2$$

$$\text{Wrong } (\bar{x}) = \frac{\Sigma x}{N}$$

$$20 = \frac{\Sigma x}{20}$$

$$\Sigma x = 600$$

$$\text{Correct } (\bar{x}) = \frac{600 - 22 + 32}{20} = \frac{610}{20} = 30.5$$

2-solo $N = 42$ students

$$(\bar{x})_w = 76$$

$\therefore$ Incorrect value = $50 + 65 = 115$

Correct value = 80

$$\therefore 76 = \frac{\sum x}{42}$$

$$76 \times 42 = \sum x$$

$$3192 = \sum x$$

$$\therefore (\bar{x})_c = \frac{3192 - 115 + 80}{42 - 1}$$

$$= \frac{3192 - 35}{41} = \frac{3157}{41} = 77$$

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